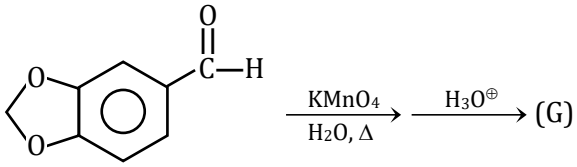
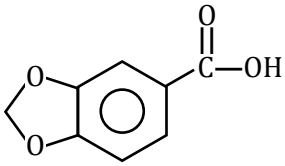
RACE
Regular Analysis Through Continuous Exercise

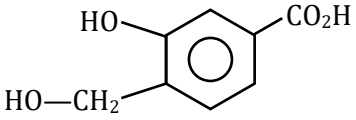
Oxidation : RACE-01

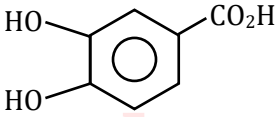
Oxidation of alcohol by Strong oxidising agent ($KMnO_4$, $K_2Cr_2O_7$, CrO_3, H_2SO_4)

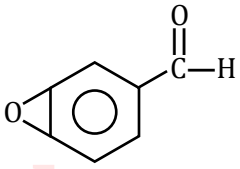
1. 

Product (G) is :

(A) 

(B) 

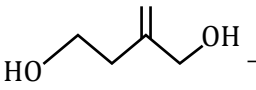
(C) 

(D) 

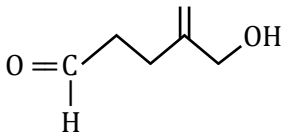
COX105

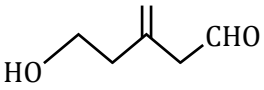
Rankers Academy JEE

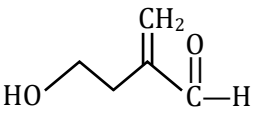
Oxidation of alcohol by MnO_2

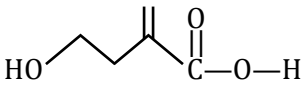
2. 

$\xrightarrow{MnO_2}$ (A), Product (A) is :

(A) 

(B) 

(C) 

(D) 

COX103

Oxidation of diol by HIO_4 and $Pb(OAc)_4$

3. **Reaction :** $CH_3 - CH = CH - CH_3 \xrightarrow[Cold]{KMnO_4}$ (A) $\xrightarrow{NaIO_4}$ (B) 2 mole

Product (*B*) is :

(A) CH_3CHO

(B) CH_3CO_2H

(C) CH_3CH_2OH

(D) $CH_3 - \underset{OH}{CH} - \underset{OH}{CH} - CH_3$

COX104

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Oxidation of Alkenes (Baeyer's Reagent, OsO_4, Hot $KMnO_4$, Acidic $KMnO_4$, Peroxy Acid followed by hydrolysis, Ag/O_2 followed by hydrolysis)

4. $CH_3–CH=CH–CH_3 \xrightarrow{X}$ Product is Y (non–resolvable) then X can be –
(cis-2-butene)

(A) Br_2 water (B) $HCO_3H/H_3O^{\oplus}$

(C) Cold alkaline $KMnO_4$ (D) All of these

COX116

5. Oxidation of isobutylene with hot acidified potassium permanganate gives:

(A) Acetone + CO_2 (B) Acetic acid

(C) Acetic acid + CO_2 (D) Acetic acid + acetone

COX112

6. During Epoxide formation, the peroxy-acid acts as an :-

(A) Nucleophile (B) Electrophile (C) Base (D) Acid

COX113

7. $CH_3–C\equiv C–CH_3 \xrightarrow[\text{S-quinoline}]{H_2,\text{Pd-CaCO}_3}$ (A) $\xrightarrow[H_2O/H^{\oplus}]{\text{m-CPBA}}$ Products

Product contains :-

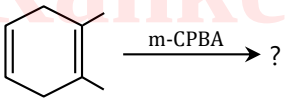
(A) Erythro Racemic (B) Threo Racemic

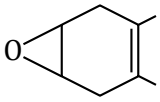
(C) Meso Compound (D) Diastereomers

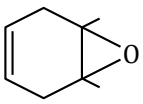
COX114

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8. Identify the major product for the following reaction :-

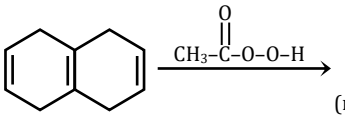


(A) 

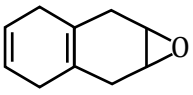
(B) 

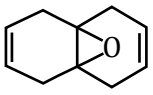
(C) Both (A) and (B) (D) None of these

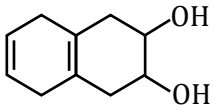
COX115

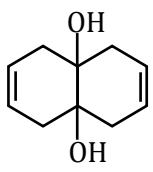
9. 

(A) ; Product (A) is :
(major)

(A) 

(B) 

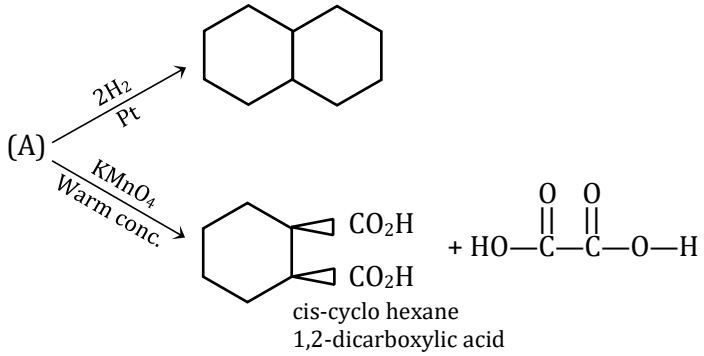
(C) 

(D) 

COX102

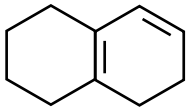
For More Join: @IITJEE_Advanced

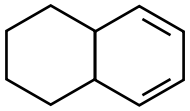
10.

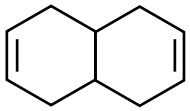


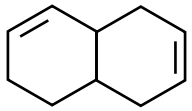
cis-cyclo hexane
1,2-dicarboxylic acid

Identify (A).

(A) 

(B) 

(C) 

(D) 

COX106

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	A	C	A	B	B	B	B	B

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